

Operations Research – Linear Programming –

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dr. ing. ir. Satish Kumar Damodar MBA, Dip.TD, IBP, CSSC

Linear programming model

Used for **optimum allocation of scarce or limited resources** to competing products

Used for solving broad range of problems arising in business, government, industry, hospitals, production, finance, marketing, research and development and personnel management for decision making.

Useful in **allocation of available resources that are limited.**

Determines optimal product mix (a combination of products, which gives maximum profit), transportation schedules, assignment problem and many more.

Product mix problem (Example)

A company manufactures two products X and Y, which require, the following resources.

The resources are the capacities machine M1, M2, and M3.

The available capacities are 50, 25, and 15 hours respectively in the planning period.

Product X requires 1 hour of machine M2 and 1 hour of machine M3.

Product Y requires 2 hours of machine M1, 2 hours of machine M2 and 1 hour of machine M3.

The profit contribution of products X and Y are \$5 and \$4 respectively.

<i>Machines</i>	<i>Products</i>		<i>Availability in hours</i>
	<i>X</i>	<i>Y</i>	
M_1	0	2	50
M_2	1	2	25
M_3	1	1	15
Profit in \$ Per unit	5	4	

Product mix problem (Example)

Products X and Y are competing variables

X and Y should not exceed available capacities

Both companies can stop the production of both x and y or can manufacture any amount of x and y

It cannot manufacture negative quantities of x and y

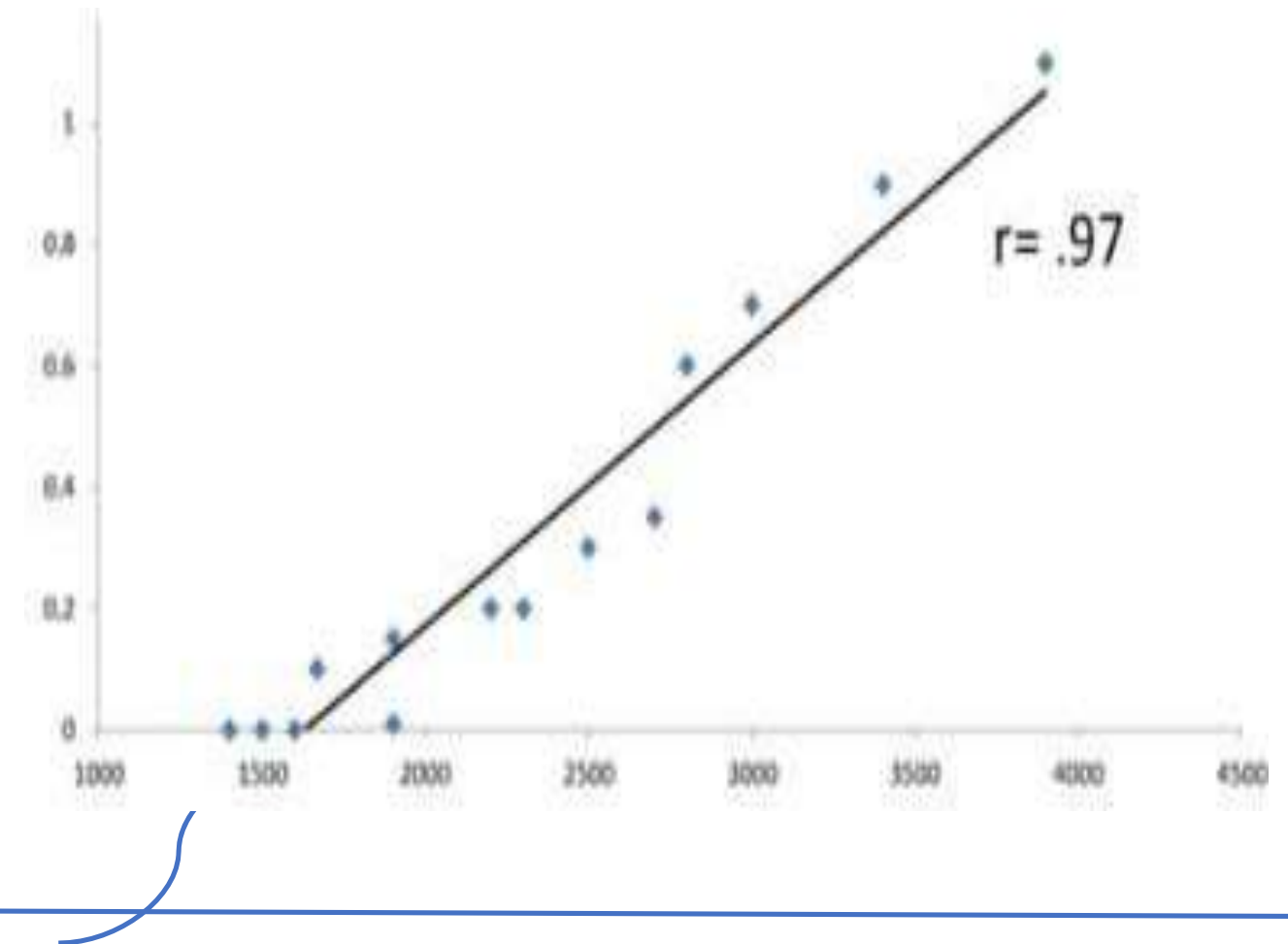
Maximize $Z = 5x + 4y$ —————> OBJECTIVE FUNCTION.

**For Machine M_1 $0x + 2y \leq 50$
For Machine M_2 $1x + 2y \leq 25$ and
For machine M_3 $1x + 1y \leq 15$ || —————> LINEAR STRUCTURAL CONSTRAINTS.**

Both x and y are ≥ 0 . —————> NON -NEGATIVITY CONSTRAINT.

Properties of linear programming model

- (a) **Linear relationship** among variables and constraints.
- b) The model must have an **objective function**.
- (c) The model must have **structural constraints**.
- (d) The model must have **non-negativity constraint**.



Linear relationship = Consistent, proportional relationship that can be graphically represented by a straight line

Basic assumptions of Linear Programming

1. The decision is **certain** (i.e., deterministic conditions) regarding all aspects of the situation;
i.e., availability of resources, profit contribution of the products, technology, courses of action and their consequences etc.
2. The relationship between variables in the problem and the resources available
i.e., constraints of the problem exhibits **linearity**.
Here the term linearity implies proportionality and additivity.
This assumption simplifies the model.
3. The technology used is fixed.
Fixed technology refers to the fact that the production requirements are fixed during the planning period
and will not change in the period.

Basic assumptions of Linear Programming

4. It is assumed that the profit contribution of a product remains constant, irrespective of level of production and sales.
5. The decision variables are continuous.

The companies manufacture products in fractional units.

For example, company manufactures 2.5 vehicles, 3.2 barrels of oil etc. This is referred to as the **assumption of divisibility**.
6. It is assumed that only one decision is required during the planning period.

This condition reveals that the linear programming model is a **static model**, which implies that linear programming problem is a single stage decision problem.

(Note: Dynamic Programming problem is a multistage decision problem).
7. All variables are restricted to non-negative values (i.e., their numerical value will be ≥ 0).

Terms used

Linear programming is a method of obtaining an optimal solution or program (say, product mix in a production problem), when resources are limited resources and there are competing candidates to consume the limited resources in certain proportion.

The term linear implies the condition of proportionality and additivity.

Program is referred as a course of action covering a specified period of time, say planning period.

The manager has to find out the best course of action in the interest of the organization.

This best course of action is termed as optimal course of action or **optimal solution to the problem.**

A program is optimal, when it maximizes or minimizes some measure or criterion of effectiveness, such as profit, sales or costs.

Terms used

Programing refers to a **systematic procedure by which a particular program or plan of action is designed**.

Programming consists of a **series of instructions and computational rules for solving a problem** that can be worked out manually or can fed into the computer.

In solving linear programing problem, a systematic method known as **Simplex method** is used.

The activity here refers to number of products or any items that utilization available resources in a certain required proportion.

The available resources may be of any nature, such as money, land, machine hours, and man-hours or materials.

These are limited in availability and are desired by the activities / products for consumption.

Mathematical expression of a problem

$$\begin{aligned}
 & \mathbf{Z = c_1 x_1 + c_2 x_2 + \dots c_n x_n} \text{ subjects to the conditions,} \longrightarrow \mathbf{OBJECTIVE FUNCTION} \\
 & \mathbf{a_{11} x_1 + a_{12} x_2 + a_{13} x_3 + \dots + a_{1j} x_j + \dots + a_{1n} x_n (\geq, =, \leq) b_1} \\
 & \mathbf{a_{21} x_1 + a_{22} x_2 + a_{23} x_3 + \dots + a_{2j} x_j + \dots + a_{2n} x_n (\geq, =, \leq) b_2} \\
 & \dots\dots\dots \\
 & \mathbf{a_{m1} x_1 + a_{m2} x_2 + a_{m3} x_3 + \dots + a_{mj} x_j + \dots + a_{mn} x_n (\geq, =, \leq) b_m} \\
 & \mathbf{and all } x_j \mathbf{ are } \geq 0 \longrightarrow \mathbf{NON NEGATIVITY CONSTRAINT.} \\
 & \mathbf{Where } j = 1, 2, 3, \dots n
 \end{aligned}$$

Structural Constraints

where a_{ij} , b_m & c_n are constants and

x_j is decision variable and should be positive

$\geq$ is generally used for minimization

$\leq$ is used for maximization

values on Right Hand Side

Steps in formulating Linear Programming problem

1. Identify the unknown decision variables to be determined and assign symbols to them.
2. Identify all restrictions or constraints in the problem and express them as a linear equation or inequalities of decision variables.
3. Identify objective and represent it as a linear function of decision variables

Maximization problem (Example 1)

A retail store stocks two types of shirts A and B in a storage box.

During a week the store can sell a maximum of 400 shirts of type A; and a maximum of 300 shirts of type B.

The storage capacity, however, is limited to a maximum of 600 of both types combined.

Type A shirt fetches a profit of \$ 2 per unit and type B a profit of \$ 5 per unit.

How many of each type the store should stock per week to maximize the total profit?

Formulate a mathematical model of the problem.

Solution (Example 1)

Shirts A and B are problem variables.

Let the store stock 'a' units of A and 'b' units of B.

As the profit contribution of A and B are \$2 and \$5 respectively,

The objective function is: Maximize $Z = 2a + 5b$ subjected to condition (s.t.)

Structural constraints are, stores can sell 400 units of shirt A; and 300 units of shirt B and storage capacity of both put together is 600 units.

The model will be:

Maximize $Z = 2a + 5b$ s.t.

$$1a + 0b \leq 400$$

$$0a + 1b \leq 300$$

$$1a + 1b \leq 600 \text{ and}$$

$$\text{Both } a \text{ and } b \text{ are } \geq 0$$

Maximization problem (Example 2)

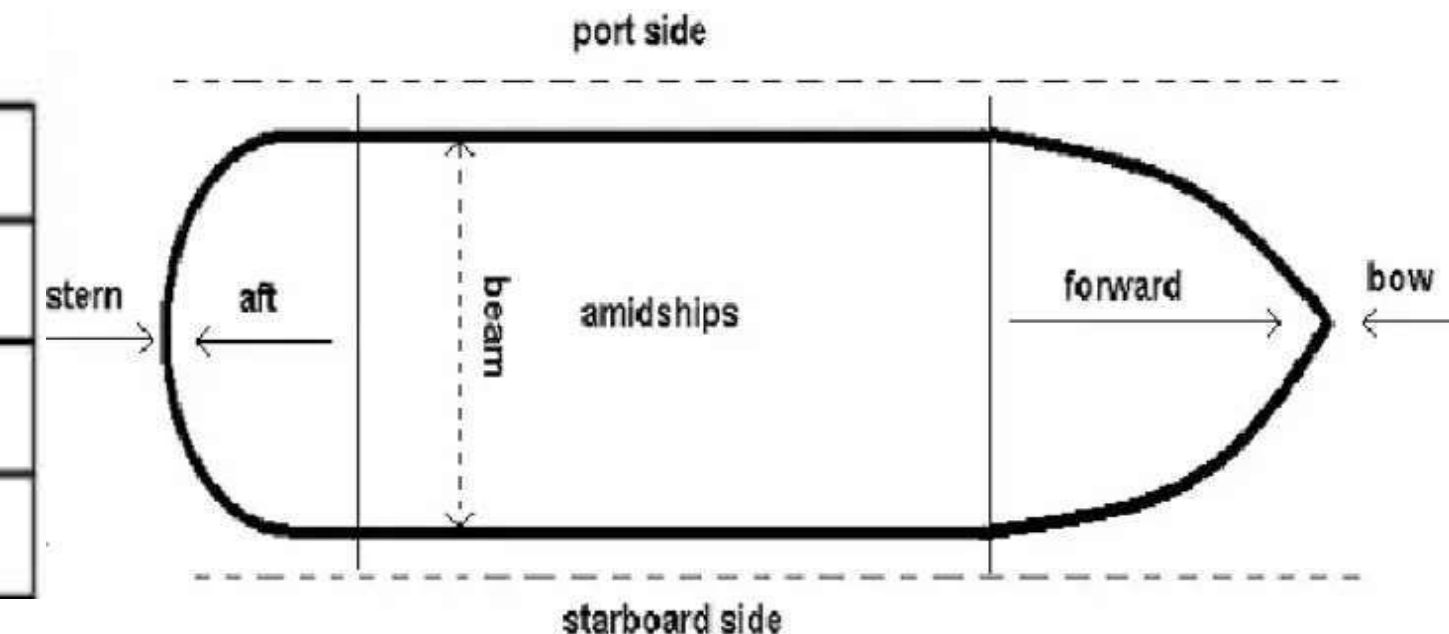
A ship has three cargo holds, forward, aft and center.

The capacity limits of the ship are:

	Weight (Tons)	Volume (Cubic meter)
Forward	2000	100,000
Centre	3000	135,000
Aft	1500	30,000
Total	6500	265,000

The following cargoes are offered, the ship owners may accept all or any part of each commodity:

<i>Commodity</i>	<i>Amount in tons.</i>	<i>Volume/ton in cubic meters</i>	<i>Profit per ton in \$</i>
A	6000	60	60
B	4000	50	80
C	2000	25	50



In order to preserve the trim of the ship the weight in each hold must be proportional to the capacity in tons.

How should the cargo be distributed so as to **maximize profit**?

Formulate this as linear programming problem

Standardizing
tonnage into volume
among the 3 commodities A, B & C

<i>Commodity</i>	<i>Amount in tons.</i>	<i>Volume/ton in cubic meters</i>
A	6000	60
B	4000	50
C	2000	25

A = 6000 tons

Volume/ton = 60m³

1 ton has a volume of 60m³

1 ton = 60 m³

1/60 ton = 1 m³

For 6000 tons, volume = $1/60 * 6000 \text{ m}^3 = 100 \text{ m}^3$

B = 4000 tons

Volume/ton = 50m³

1 ton has a volume of 50 m³

1 ton = 50m³

1/50 ton = 1m³

For 4000 tons, volume = $1/50 * 4000 \text{ m}^3 = 80 \text{ m}^3$

C = 2000 tons

Volume/ton = 25m³

1 ton has a volume of 25 m³

1 ton = 25m³

1/25 ton = 1m³

For 2000 tons, volume = $1/25 * 2000 \text{ m}^3 = 80 \text{ m}^3$

	Weight (Tons)	Volume (Cubic meter)
Forward	2000	100,000
Centre	3000	135,000
Aft	1500	30,000
Total	6500	265,000

Commodities are A, B & C with respective units a, b & c

The objective is to maximize profits

Then, **Objective function** is: **Maximize $Z = 60a + 80b + 50c$ s.t.**

Constraints: Weight (Tonnage) and Volume (Cubic meter)

Weight constraint: **$6000a + 4000b + 2000c \leq 6500$**

The tonnage of commodity is 6000 and each ton occupies 60 cubic meters,

hence there are 100 cubic meters capacity is available for A

Volume constraint: $100a + 80b + 80c \leq 2,65,000$

Maximise $Z = 60a + 80b + 50c$ s.t.

$6000a + 4000b + 2000c \leq 6,500$

$100a + 80b + 80c \leq 2,65,000$ and

$a, b, c \geq 0$

Minimization problem (Example 1)

A doctor examines a patient and find out that and he is deficient in two vitamins- vitamin A and vitamin D.

The doctor prescribes tonic X and tonic Y, containing vitamin A, and vitamin D in certain proportion.

Also advises the patient to consume **at least** 40 units of vitamin A and 50 units of vitamin D everyday.

The cost of tonics X and Y and the proportion of vitamin A and D that present in X and Y are given in the table below.

<i>Vitamins</i>	<i>Tonics</i>		<i>Daily requirement in units.</i>
	<i>X</i>	<i>Y</i>	
<i>A</i>	2	4	40
<i>D</i>	3	2	50
Cost in \$ per unit.	5	3	

Formulate linear program to minimize the cost of tonics

Solution

Let patient purchase x units of X and y units of Y .

Objective function: **Minimize $Z = 5x + 3y$**

Inequality for vitamin A is **$2x + 4y \geq 40$ (at least 40 units of A)**

(Here at least word indicates that the patient can consume more than 40 units but not less than 40 units of vitamin A daily).

Similarly the inequality for vitamin D is **$3x + 2y \geq 50$ (At least 50 units of B)**

For non-negativity constraint the patient cannot consume negative units.

Hence both **x and y must be ≥ 0**

Model:

Minimize $Z = 5x + 3y$ s.t.

$2x + 4y \geq 40$

$3x + 2y \geq 50$ and

Both x and y are ≥ 0 .

Vitamins	Tonics		Daily requirement in units.
	X	Y	
A	2	4	40
D	3	2	50
Cost in \$ per unit.	5	3	

Methods for solving linear programming problem

The Graphical Method:

When there are two decision variables in the problem.

The **graphical method solves only two variable** problems.

1. The Systematic Trial and Error method:

In this method, values are given to variables until optimal solution is achieved.

This method is time consuming and laborious, and not discussed.

2. The Vector method:

In this method each decision variable is considered as a vector and principles of vector algebra is used to obtain optimal solution.

This method is also time consuming, and not discussed.

3. The Simplex method:

When there are **more than two decision variables**, **simplex method** is used to solve the problem. It is systematic program.

Problem with two variables is solved using both graphical and simplex method, to understand relationship between the two variables.

Graphical method

Inequalities (structural constraints) are equations and cannot be represented graphically

Only **two variable problems** are considered, because straight lines are represented in two-dimensional plane (X- axis and Y-axis).

Nonnegativity constraints (decision variables) must have positive values always, and **the solution to the problem lies in FIRST QUADRANT** of the graph.

Some times the **value of variables may fall in other quadrants**.

In such cases, the **line joining the values of the variables must be extended to the first quadrant**.

Graphical problem 1

A company manufactures two products, X and Y by using three machines A, B, and C. Machine A has 4 hours of capacity available during the coming week. The available capacity of machines B and C during the coming week is 24 hours and 35 hours respectively. One unit of product X requires one hour of Machine A, 3 hours of machine B and 10 hours of machine C. One unit of product Y requires 1 hour, 8 hour and 7 hours of machine A, B and C respectively. When one unit of X is sold in the market, it yields a profit of \$ 5 per unit and that of Y is \$ 7 per unit. Solve the problem by using graphical method to find the optimal product mix

<i>Machines</i>	<i>Products</i> <i>(Time required in hours).</i>		<i>Available capacity in hours.</i>
	<i>X</i>	<i>Y</i>	
<i>A</i>	1	1	4
<i>B</i>	3	8	24
<i>C</i>	10	7	35
Profit per unit in \$	5	7	

Solution

Let the company manufactures x units of X and y units of Y .

The L.P. model is:

Maximize $Z = 5x + 7y$ s.t. (subject to)

$$1x + 1y \leq 4$$

$$3x + 8y \leq 24$$

$$10x + 7y \leq 35 \text{ and}$$

Both x and y are ≥ 0 .

As we cannot draw graph for inequalities, consider them as equations.

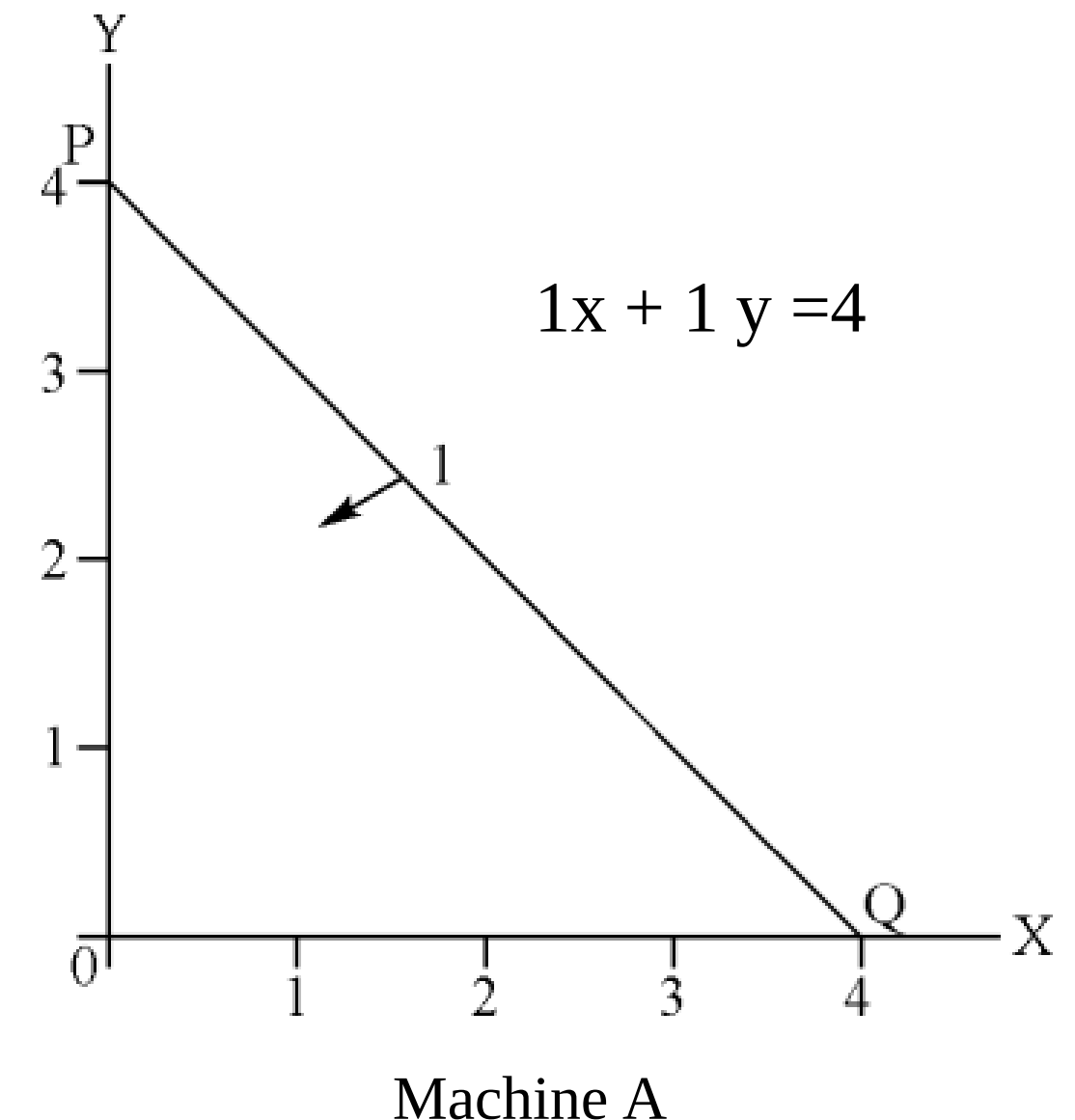
Let us take machine A. and find the boundary conditions.

If $x = 0$, machine A can manufacture $4/1 = 4$ units of y .

Similarly, if $y = 0$, machine A can manufacture $4/1 = 4$ units of x .

P0Q represents capacity of machine A

Machines	Products (Time required in hours).		Available capacity in hours.
	X	Y	
A	1	1	4
B	3	8	24
C	10	7	35
Profit per unit in \$	5	7	

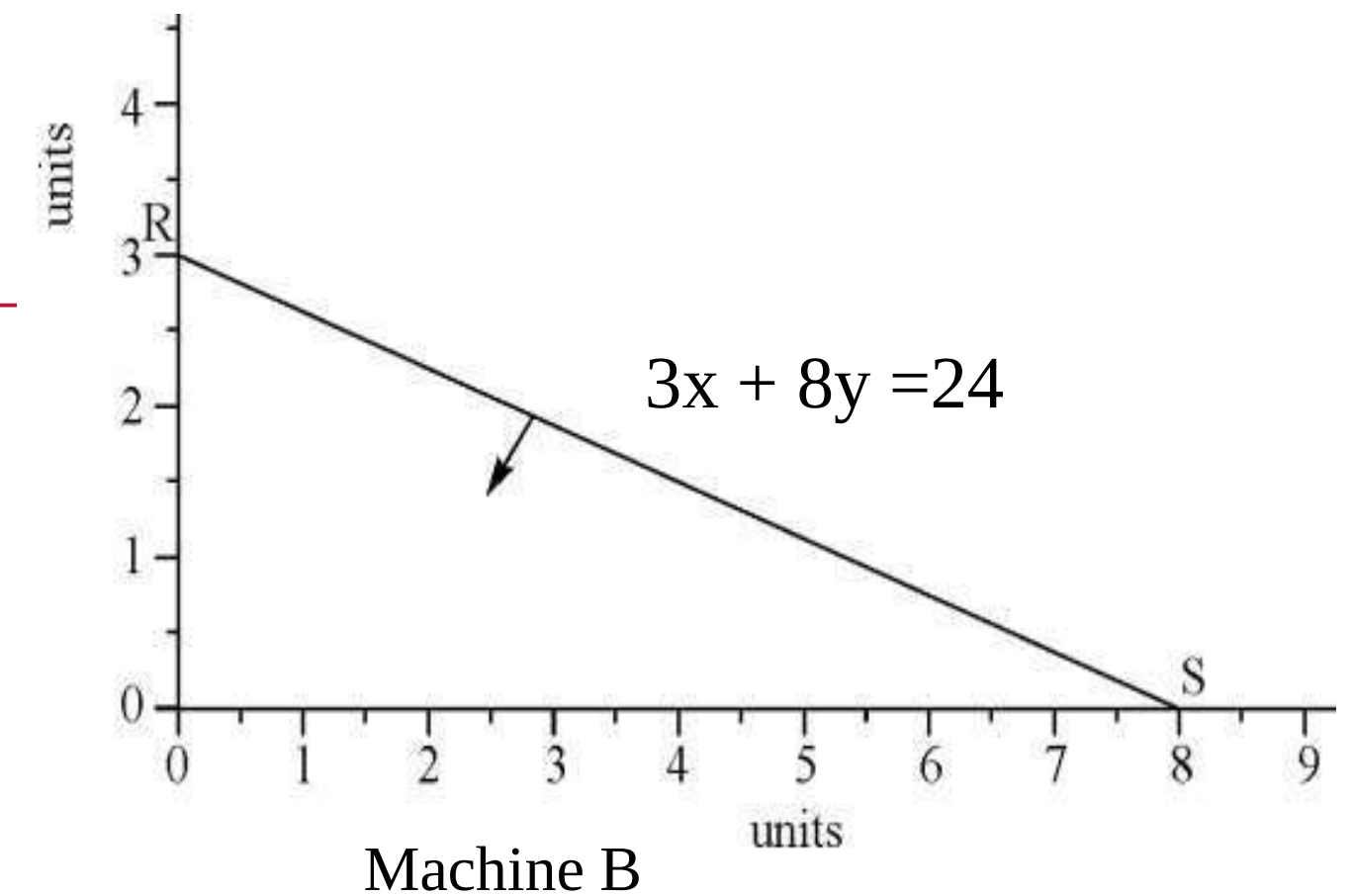


Solution

Consider machine B

When $x = 0$, $y = 24/8 = 3$ and when $y = 0$ $x = 24/3 = 8$

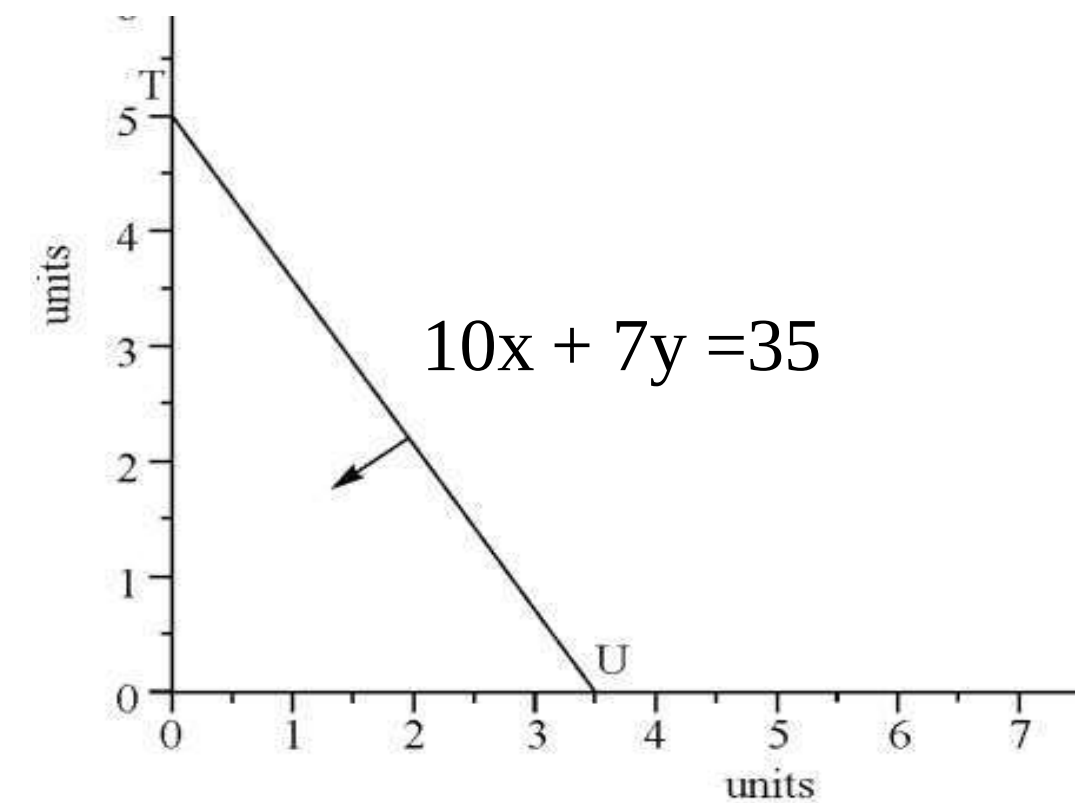
R0S represents capacity of machine B



Consider machine C

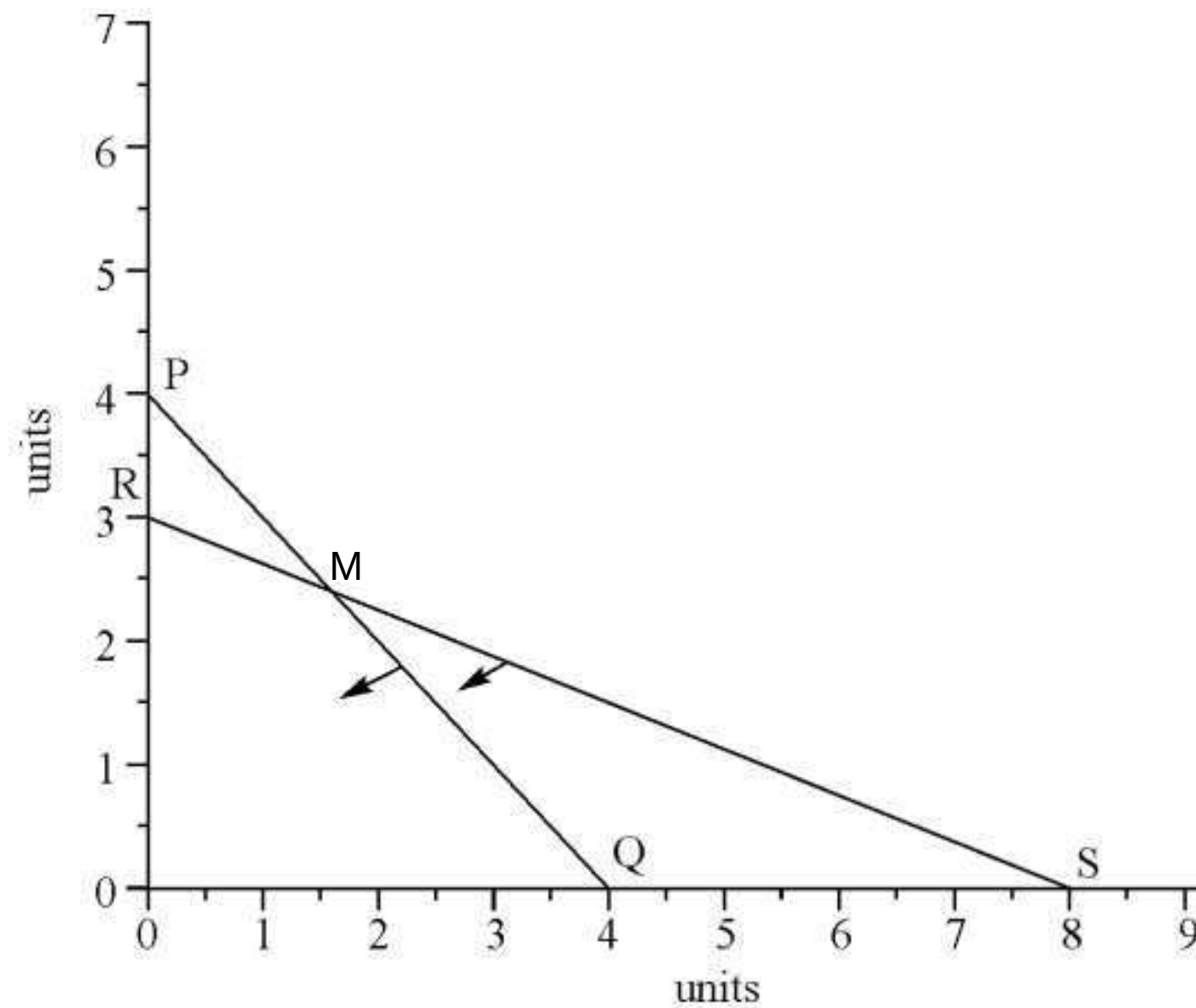
When $x = 0$, $y = 35/10 = 3.5$ and when $y = 0$, $x = 35 / 7 = 5$

T0U represents capacity of machine C

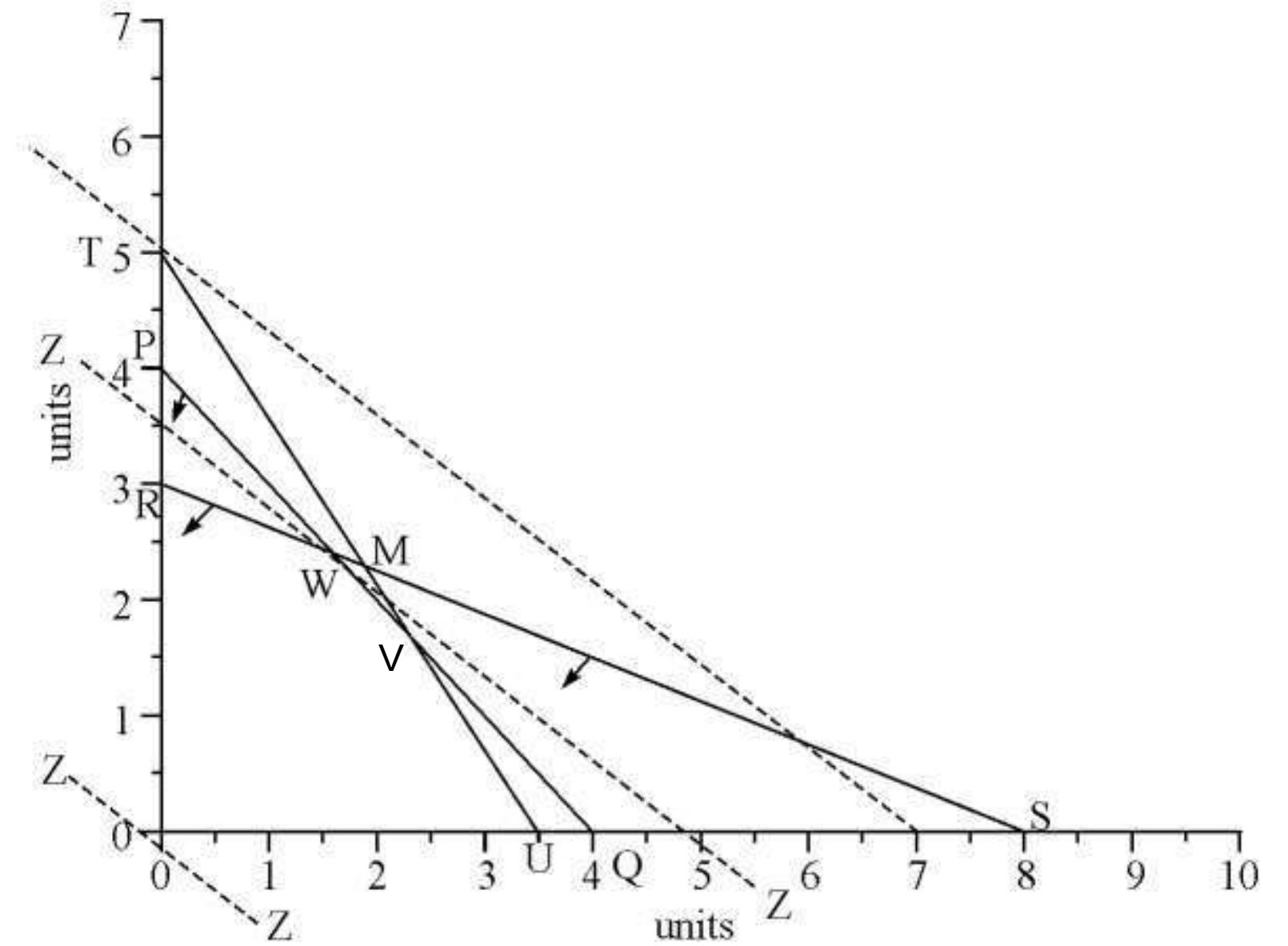


Machine C

Solution



Machines A & B



Machines A, B & C combined

Solution

The combined graphs of machines A and B meets at M

The three machines A, B and C represents a polygon R0UVW

For maximization problem, we select coordinates that gives maximum value For minimization problem, we select coordinates that gives minimum value

The coordinates are

$R = (0, 3)$, $O = (0, 0)$, $U = (3.5, 0)$, $V = (2.5, 1.5)$ and $W = (1.6, 2.4)$.

Substituting these values in objective function $5x + 7y$ we get

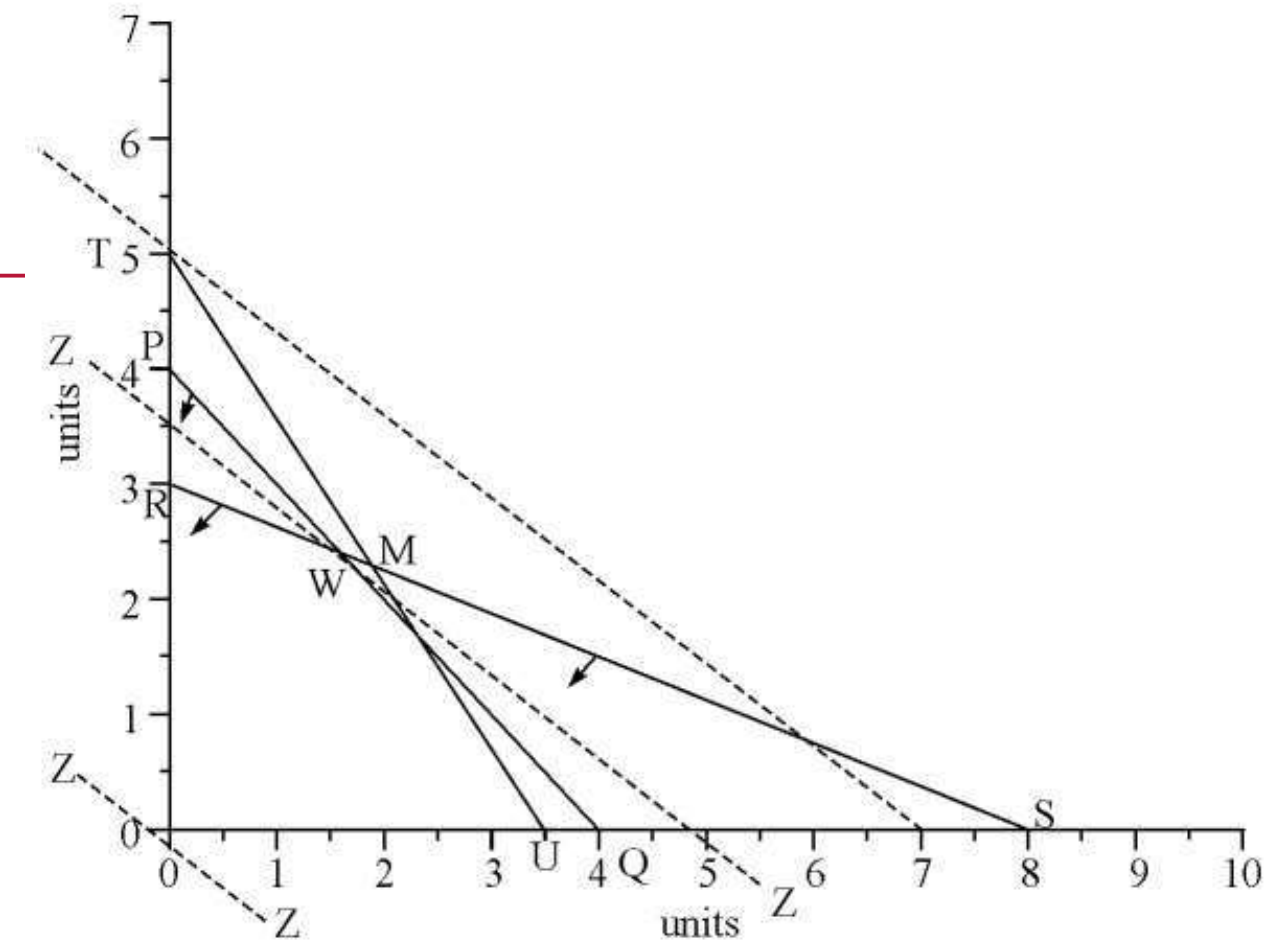
$$Z(0, 3) = 5 \times 0 + 7 \times 3 = \$ 21.00, \text{ at point R}$$

$$Z(0, 0) = 5 \times 0 + 7 \times 0 = \$ 00.00, \text{ at point O}$$

$$Z(3.5, 0) = 5 \times 3.5 + 7 \times 0 = \$ 17.5 \text{ at point U}$$

$$Z(2.5, 1.5) = 5 \times 2.5 + 7 \times 1.5 = \$ 23.00 \text{ at point V}$$

$$Z(1.6, 2.4) = 5 \times 1.6 + 7 \times 2.4 = \$ 24.80 \text{ at point W is the optimal solution (Z Maximization)}$$



Machines A, B & C combined

Graphical problem 2

A company produces two types of items P and Q that require gold and silver.

Each unit of type P requires 4g silver and 1g gold while that of type Q requires 1g silver and 3g gold.

The company produces 8g silver and 9g gold.

If each unit of type P brings a profit of \$44 and that of type Q brings a profit of \$55, determine the number of units of each type that the company should produce to maximize the profit.

What is the maximum profit?

Solution

Let x be the quantity of type P to be produced and y be the quantity of type Q to be produced.

	Silver	Gold	Profit
Type P	4g	1g	\$ 44 per unit
Type Q	1g	3g	\$ 55 per unit
Available (at the most)	8g	9g	

Maximize $Z = 44x + 55y$ subject to the constraints:

$$4x + y \leq 8$$

$$x + 3y \leq 9$$

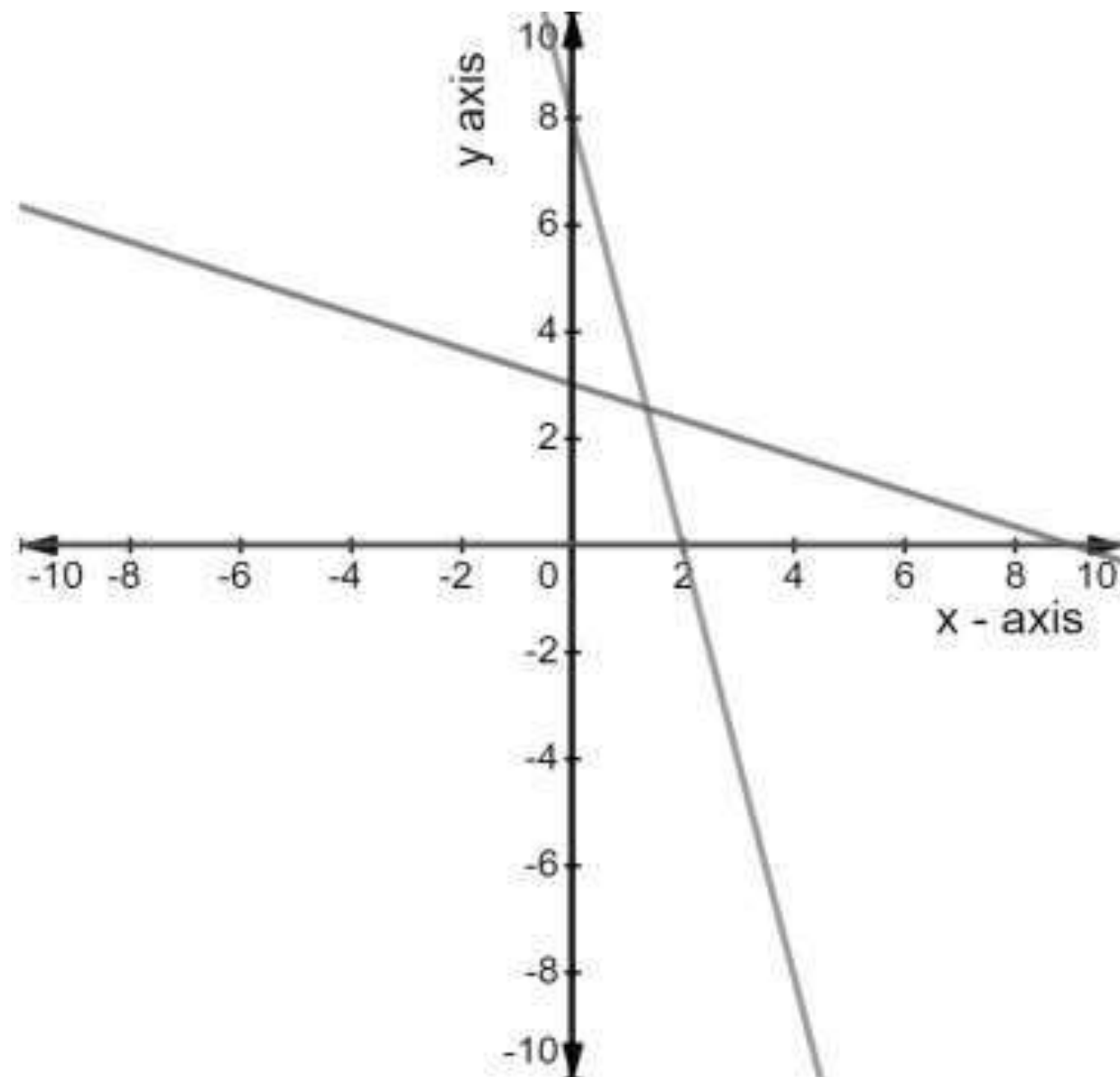
$$\text{and } x \geq 0, y \geq 0.$$

Solution

Plotting graphs for equations $4x + y = 8$, $x + 3y = 9$, $x = 0$, $y = 0$

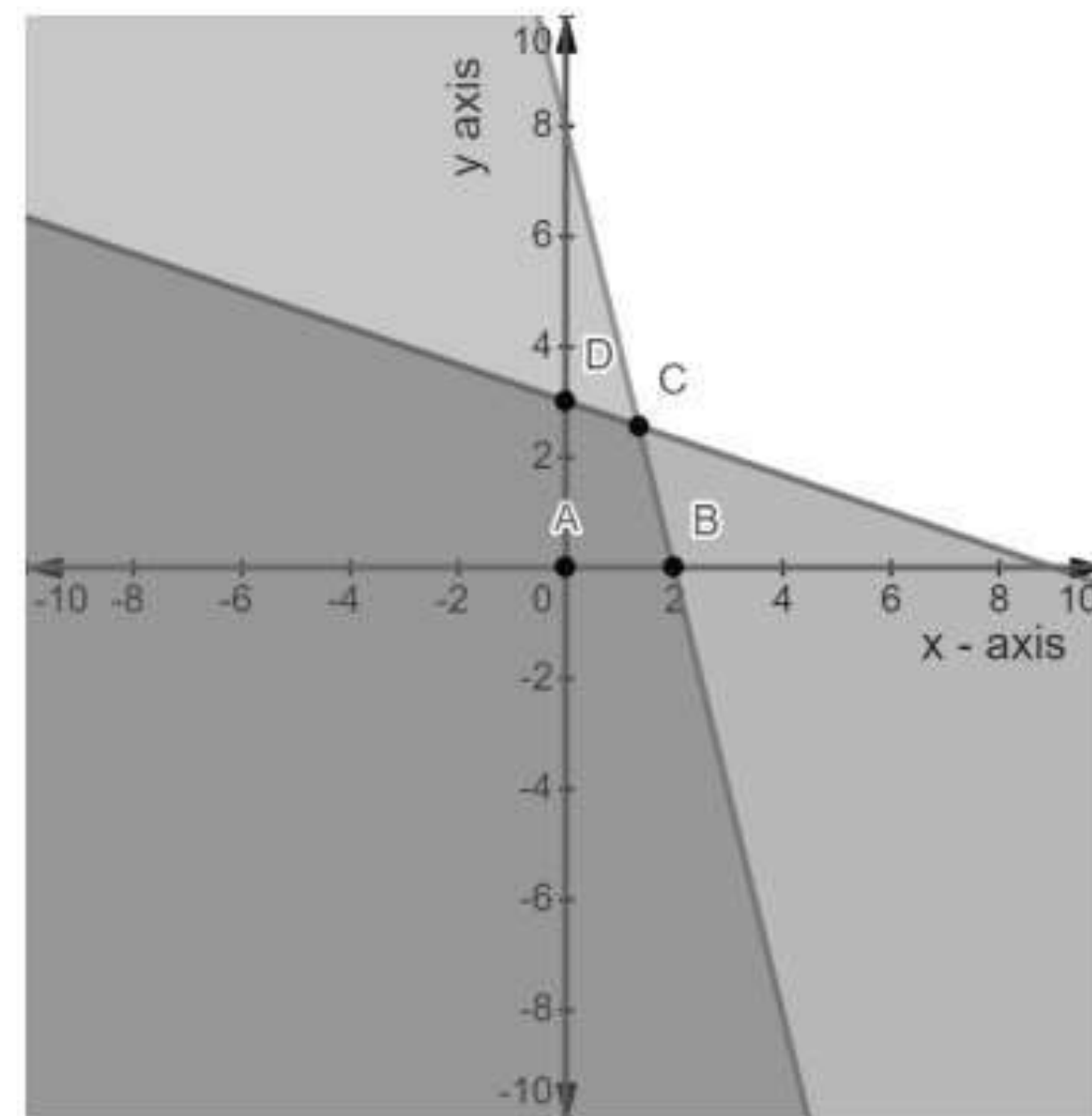
For the line $4x + y = 8$

x	0	2
y	8	0



For the line $x + 3y = 9$

x	0	9
y	3	0



Solution

$$\text{Maximum } Z = 44x + 55y$$

From the graph, the points are

$$Z(0, 0) = 44(0) + 55(0) = 0$$

$$Z(2, 0) = 44(2) + 55(0) = 88$$

$$\mathbf{Z(15/11, 28/11) = 44(15/11) + 55(28/11) = 200}$$

$$Z(0, 3) = 44(0) + 55(3) = 165$$

The value of Z is maximum at $C(15/11, 28/11)$ and optimum solution is Max. $Z = 200$ when $x = 15/11$ and $y = 28/11$

The value of x and y at point C on graph is obtained by solving both equations

$$4x + y = 8$$

$$x + 3y = 9$$

$$0 + 11y = 28$$

$$y = 28/11$$

Substituting the value of y in the equation, we get $x = 15/11$

$$C = (15/11, 28/11)$$

Assignment 1

1. Maximize $Z = 6X + 3Y$ subjects to the constraints $2X + 5Y \leq 120$, $4X + 2Y \leq 80$, $X \geq 0$, $Y \geq 0$.
2. Maximize $Z = 3x_1 + 2x_2$ subjects to the constraints $x_1 - x_2 \leq 1$, $x_1 + x_2 \geq 3$, $x_1 \geq 0$, $x_2 \geq 0$
3. Maximize $Z = x_1 + x_2$ subjects to the constraints $x_1 + x_2 \leq 1$, $-3x_1 + x_2 \geq 3$, $x_1 \geq 0$, $x_2 \geq 0$
4. A company manufactures two products X and Y, each of which requires three types of processing. The length of time for processing each unit and the profit per unit are given in the following table:

	Product X (hr/unit)	Product Y (hr/unit)	Available capacity per day (hr)
Process 1	12	12	840
Process 2	3	6	300
Process 3	8	4	480
Profit per unit	5	7	

How many units of each product should the company manufacture per day in order to maximize profit?

Expressing LPP in Canonical form

- (a) Objective function should be of **maximization** form.

If it is given in minimization form, it should be converted into maximization form.

- (b) All the constraints should be of “ $\leq$ ” type, except for non- negative restrictions.

Inequality of “ $\geq$ ” type, if any, should be changed to an inequality of the “ $\leq$ ” type

- (b) All variables should be **non-negative**.

If a given variable is **unrestricted** in sign (i.e., positive, negative or zero), it can be written as a difference of two non-negative variables.

Suppose x is unrestricted in sign, then x can be written as $x = x' - x''$ where $x' \geq 0, x'' \geq 0$.

Expressing LPP in Canonical form

Minimize $Z = 2x_1 + x_2 + 4x_3$ Subject to constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$x_1 + 2x_2 + x_3 \geq 5$$

$$2x_1 + 3x_3 \leq 2$$

$x_1, x_2 \geq 0$ and x_3 is unrestricted in sign ($x_3 > 0, x_3 < 0$)

Here, the objective function is of the minimization form.

We rewrite the maximization form as:

$$\text{Minimize } Z = 2x_1 + x_2 + 4x_3$$

We have to maximize $-Z = -2x_1 - x_2 - 4x_3$.

The problem becomes,

Maximize $Z' = -2x_1 - x_2 - 4x_3$. where $Z' = -Z$

The second constraints is of the type “ $\geq$ ”.

Convert it into type “ $\leq$ ”, we multiply the inequality by -1 and write $-x_1 - 2x_2 - x_3 \leq -5$

Other constraints are already in the desired form. But **x_3 is unrestricted** in sign.

So, we write **$x_3 = x_3' - x_3''$** ,

where $x_3' \geq 0, x_3'' \geq 0$

The canonical form of the given problem is

Maximize $Z' = -2x_1 - x_2 - 4(x_3' - x_3'')$, where $Z' = -Z$

Subject to the constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$-x_1 - 2x_2 - (x_3' - x_3'') \leq -5$$

$$2x_1 + 3(x_3' - x_3'') \leq 2$$

$$x_1 \geq 0, x_2 \geq 0, x_3' \geq 0, x_3'' \geq 0$$

Expressing LPP in Standard form

- 1) The objective function should be in the **maximization** form.
- 2) The **right-side element of each constraint should be non- negative**.
If it is negative, we multiply the inequality by -1.
- 3) All constraints should be **expressed in the form of equations**, except for the non- negative restrictions
- 4) Augmenting a **slack or surplus variables** to the function and balance it by adding the slack variables to the objective function

Expressing LPP in Standard form

Minimize $Z = 2x_1 + x_2 + 4x_3$ Subject to the constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$x_1 + 2x_2 + x_3 \geq 5$$

$$2x_1 + 3x_3 \leq -2$$

$$x_1 \geq 0, x_2 \geq 0 \text{ and } x_3 \geq 0$$

Here, the objective function is of the minimization form.

We rewrite it in the maximization form as follows:

$$\text{Minimize } Z = 2x_1 + x_2 + 4x_3$$

We have to maximize $-Z = -2x_1 - x_2 - 4x_3$.

The problem becomes,

Maximize $Z' = -2x_1 - x_2 - 4x_3$. where $Z' = -Z$

The inequalities are to be converted to equations.

The first and third inequalities are of the type “less than or equal to ($\leq$)”.

A slack variable is added to the left side of each of the first and third inequalities.

The second inequality is of the type “more than or equal to ($\geq$)”.

So, a surplus variable is to be subtracted from the left side of this inequality.

Thus, a standard form of the given LPP is

Maximize $Z' = -2x_1 - x_2 - 4x_3 + 0S_1 + 0S_2 + 0S_3$,
where $Z' = -Z$ subject to constraints:

$$-2x_1 + 4x_2 + s_1 = 4$$

$$-x_1 - 2x_2 - x_3 - s_2 = -5$$

$$2x_1 + 3x_3 + s_3 = -2$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, s_1 \geq 0, s_2 \geq 0, s_3 \geq 0$$

Canonical form & Standard form

Feature	Canonical Form	Standard Form
Objective Function	Usually Maximization	Can be Maximization or Minimization
Constraints	All are Inequalities (typically $\leq$)	All are strict Equalities ($=$)
Variables	All decision variables must be ≥ 0	All variables (including slack/surplus) must be ≥ 0
Right-Hand Side (b)	Can be positive or negative	Must be non-negative ($b \geq 0$)
Primary Use Case	Duality theory and economic interpretation	Inputting into the Simplex Algorithm

Problem

A machine tool company conducts a job-training at a ratio of one for every ten trainees.

The training lasts for one month.

From past experience it has been found that out of 10 trainees hired, only seven complete the program successfully.

(The unsuccessful trainees are released). Trained machinists are also needed for machining.

The company's requirement for the next three months is as follows: January: 100 machinists, February: 150 machinists and March: 200 machinists. In addition, the company requires 250 trained machinists by April.

There are 130 trained machinists available at the beginning of the year. Pay roll cost per month is: Each trainee \$ 400 per month.

Each trained machinist (machining or teaching): \$ 700 p.m.

Trained machinist who is idle: \$ 500 p.m. (Labour union forbids ousting trained machinists). Build linear program for produce the minimum cost hiring and training schedule and meet the company's requirement.

Solution

There are three options for trained machinists as per the data given.

- (i) A trained machinist can work on machine,
- (ii) he can teach or
- (iii) he can remain idle.

It is given that the number of trained machinists available for machining is fixed.

The unknown decision variables are the number of machinists goes for teaching and those who remain idle for each month.

Let,

- ‘a’ be the trained machinists teaching in the month of January.
- ‘b’ be the trained machinists idle in the month of January.
- ‘c’ be the trained machinists for teaching in the month of February.
- ‘d’ be the trained machinists remain idle in February.
- ‘e’ be the trained machinists for teaching in March.
- ‘f’ be the trained machinists remain idle in the month of March.

Solution

The constraints can be formulated by the rule that the number of machinists employed (machining + teaching + idle) = Number of trained machinists available at the beginning of the month.

For January $100 + 1a + 1b \geq 130$

For February, $150 + 1c + 1d = 130 + 7a$

(Here $7a$ indicates that the number of machinist trained is $10 \times a = 10a$.
But only 7 of them are successfully completed the training i.e. $7a$).

For the month of March, $200 + 1e + 1f \geq 130 + 7a + 7c$

The requirement of trained machinists in the month of April is 250, the constraints for this will be

$130 + 7a + 7c + 7e \geq 250$ and

The objective function is

Minimize $Z = 400 (10a + 10c + 10e) + 700 (1a + 1c + 1e) + 500 (1b + 1d + 1f)$ and

the nonnegativity constraint is $a, b, c, d, e, f \geq 0$.

Solution

The model is:

Minimize $Z = 400 (10a + 10c + 10e) + 700 (1a + 1c + 1e) + 500 (1b + 1d + 1f)$ s.t.

$$100 + 1a + 1b \geq 130$$

$$150 + 1c + 1d \geq 130 + 7a$$

$$200 + 1e + 1f \geq 130 + 7a + 7c$$

$$130 + 7a + 7c + 7e \geq 250 \text{ and}$$

$$a, b, c, d, e, f \text{ all } \geq 0.$$

Thank You!